

Glossary of Key Terms

Deep Learning for Beginners: Sequence Models

AI Foundations Series — Book 5

This glossary collects the main terms and concepts from Book 5 — the journey from recurrent networks through word embeddings, attention, and the transformer — arranged alphabetically for quick reference. Each definition reflects how the term is used in the book itself: plain language first, with just enough precision to be useful when reviewing a chapter or returning to a companion script. Beginners can read it straight through; reviewers can jump to any entry.

A

Attention

A mechanism that lets a model look back over all of an input's positions and focus on the ones that matter for the current step, rather than relying on a single summary vector. It computes a weighted blend of the inputs, where the weights say how much to attend to each.

Attention Map

A heatmap of a model's attention weights, with output positions on one axis and input positions on the other. It shows, cell by cell, which inputs the model looked at while producing each output - making attention interpretable.

Attention Weights

The set of numbers, produced by softmax and summing to one, that say how much the model focuses on each input position at a given step. Large weight means “look here.”

B

Backpropagation Through Time (BPTT)

The training method for RNNs: unroll the network across the time steps of a sequence, then apply ordinary backpropagation through the unrolled chain to update the shared weights.

Beam Search

A decoding method that keeps the best B partial sequences (the beam width) at every step instead of committing to one. It searches more of the space than greedy decoding and usually finds a better overall sequence.

Beam Width (B)

The number of partial sequences beam search keeps alive at each step. $B = 1$ is greedy decoding; larger B searches more thoroughly at the cost of more computation.

Bidirectional RNN

An RNN that reads a sequence both forwards and backwards and combines the two, so each position's representation reflects context from both sides.

BIO Tagging

A labeling scheme for tasks like named entity recognition: B- marks the beginning of an entity, I- marks a token inside/continuing it, and O marks a token outside any entity (e.g. B-PER I-PER for a two-word person's name).

BLEU Score

An automatic measure of machine-translation quality based on n-gram overlap with human reference translations, with a clipping rule and a brevity penalty. It rewards word overlap, not meaning, so it is a useful proxy rather than a perfect judge.

C

Context Vector

The single vector an attention step produces for a given output position: a weighted sum of the encoder states, tilted toward whatever the decoder needs at that moment.

CTC (Connectionist Temporal Classification)

A method for training on unsegmented sequences (like speech) where the input is far longer than the output. The network emits one symbol per input step, including a blank, and repeated symbols not separated by a blank are collapsed to produce the transcript.

D

Decoder

The half of a sequence-to-sequence model that generates the output sequence one step at a time, conditioned on the encoder's representation of the input.

E

Embedding

A dense vector that represents a word (or token) so that similar meanings sit near each other in vector space. Pretrained embeddings let models learn from far less data. See also GloVe, Word2Vec.

Encoder

The half of a sequence-to-sequence model that reads the input sequence and compresses it into a representation (a single thought vector, or a set of states for attention to use).

Encoder-Decoder

An architecture that maps one sequence to another: an encoder reads the input into a representation, and a decoder generates the output from it. Used for translation, summarization, and image captioning.

Exploding Gradients

When gradients grow uncontrollably large during backpropagation through time, destabilizing training. Commonly mitigated with gradient clipping. Contrast with vanishing gradients.

G

GloVe (Global Vectors)

A set of pretrained word embeddings learned from global word co-occurrence statistics. Widely reused as a strong starting point for text models.

Greedy Decoding

Generating an output sequence by always taking the single most likely next token at each step. Fast but short-sighted: a locally best choice can lead to a worse overall sequence.

GRU (Gated Recurrent Unit)

A gated RNN cell that uses update and reset gates to control what information is kept or discarded over time. Simpler than an LSTM, and often comparable in performance.

H

Hidden State

The vector an RNN carries from one time step to the next, serving as its memory of everything relevant it has read so far.

L

Layer Normalization

A technique that rescales the values within a layer to a stable mean and spread, keeping numbers well-behaved. A standard “bookkeeping” component inside transformer blocks.

Length Normalization

A correction applied in beam search that divides a sequence's score by its length (raised to a power), so the search does not unfairly favor short sequences just because they multiply in fewer probabilities.

LSTM (Long Short-Term Memory)

A gated RNN cell with a separate cell state and input, forget, and output gates. Its gating lets it carry information across many time steps, addressing the vanishing-gradient problem of plain RNNs.

M

Multi-Head Attention

Running several attention mechanisms (heads) in parallel, each with its own learned projections, so different heads can capture different relationships. Their outputs are concatenated and projected into one result.

N

Named Entity Recognition (NER)

The task of finding and labeling the real-world entities in text - people, organizations, locations - framed as classifying every token with a BIO tag.

P

Positional Encoding

A pattern added to each token's vector to encode its position, since self-attention alone is order-blind. The transformer uses fixed sine and cosine waves of many wavelengths so nearby positions get similar encodings.

Q

Query, Key, Value

The three vectors each token produces in self-attention. A token's query is matched against every token's key to get attention weights; those weights then blend the values into the token's new representation.

R

Recurrent Neural Network (RNN)

A network for sequences that processes one element at a time, carrying a hidden state forward as memory. The foundation of the sequence models in this book, later surpassed by attention and transformers.

S

Self-Attention

Attention applied within a single sequence, letting every token attend to every other token in that same sequence to build a context-aware representation of itself. The core operation of the transformer.

Sequence-to-Sequence (Seq2Seq)

A model that maps an input sequence to an output sequence, often of a different length, using the encoder-decoder architecture.

Spectrogram

A picture of sound with time on one axis and frequency on the other, and brightness showing energy. It is the model-ready representation of audio used in speech and trigger-word tasks.

Subword Tokenization

Splitting text into word pieces rather than whole words, so any word - even an unseen one - can be spelled out from a fixed vocabulary of pieces. Solves the out-of-vocabulary problem; the “###” prefix marks a continuation piece.

T

Teacher Forcing

A training technique for decoders where the true previous token (rather than the model's own prediction) is fed in at each step, which speeds up and stabilizes learning.

Thought Vector

The single fixed-length vector an encoder produces to summarize an entire input sequence. For long sequences it becomes a bottleneck, which attention was invented to relieve.

Token

The basic unit a model reads - a word, subword piece, or character - after the tokenizer converts raw text into a sequence of them.

Token Classification

Predicting a label for every token in a sequence (as opposed to one label for the whole sequence). Named entity recognition is the classic example.

Tokenizer

The component that converts raw text into tokens (and their integer ids) for a model, and back again. Every pretrained model has a matching tokenizer that must be used with it.

Transformer

An architecture built entirely from attention, with no recurrence, that processes a whole sequence in parallel. Introduced in “Attention Is All You Need” (2017) and the basis of modern large language models.

Trigger Word Detection

Detecting a wake word (like “activate,” “Alexa,” or “Hey Siri”) in a stream of audio, framed as labeling each output time step 0 or 1 for “did the wake word just occur?”

V

Vanishing Gradients

When gradients shrink toward zero as they propagate back through many time steps, so early steps barely learn. The main weakness of plain RNNs, addressed by GRUs and LSTMs.

W

Word Embedding

See Embedding - a learned dense vector representation of a word that captures meaning and relationships.

Word2Vec

A method for learning word embeddings from raw text by predicting nearby words (skip-gram) or a word from its neighbors (CBOW). A foundational technique for word vectors.

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